BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY WITH HONOURS [B.Sc. IT (Hons)]

Bachelor of Science in Information Technology with Honours consists of **SEVEN (7)** components as follows:

COMPONENTS	CREDIT HOURS
A. COMPULSORY COURSES	24
B. CORE COMPUTING COURSES	18
C. DISCIPLINE CORE	36
D. FIELD ELECTIVE	18
E. FREE ELECTIVE	6
F. FINAL YEAR PROJECT	6
G. INDUSTRIAL TRAINING	12
TOTAL	120

To be awarded Bachelor of Science in Information Technology with Honours, a student is required to take and pass a minimum of 120 credit hours according to the following structure:

A. COMPULSORY COURSES (24 CREDIT HOURS)

Students must take ALL courses under compulsory courses as follows:

i. Malaysian Students

CODE	COURSE	CREDIT HOURS	CATEGORY
MPUK3112	Falsafah dan Cabaran Semasa	2	UNIVERSITY CORE COURSES
MPUK3122	Malaysian Nation-State Aspiration	2	
MPUK3312	Nationhood Studies	2	
MPUK3212	Principles of Entrepreneurship	2	
MPUK3152	Integrity And Anti-Corruption	2	
MPUK3411	Bahasa Melayu Ilmiah *audit course	1	
VXXXXXXX	Co-Curriculum	2	
BWFFK1013	Fundamentals of Finance	3	GENERAL/HEP COURSES
SQQSK1013	Elementary Statistics	3	
MPB***3	English Core 1	3	ENGLISH CORE COURSES
MPB***3	English Core 2 Pre- requisite: MPB***3 English Core 1	3	
TOTAL		**25**	

ii. International Students

CODE	COURSE	CREDIT HOURS	CATEGORY
MPUK3142	Malaysian CITRA	2	UNIVERSITY CORE COURSES
MPUK3132	Basic Malay for Foreigner	2	
MPUK3312	Nationhood Studies	2	
MPUK3212	Principles of Entrepreneurship	2	
MPUK3152	Integrity And Anti-Corruption	2	
VXXXXXXX	Co-Curriculum	2	
BWFFK1013	Fundamentals of Finance	3	GENERAL/HEP COURSES
SQQSK1013	Elementary Statistics	3	
MPB***3	English Core 1	3	ENGLISH CORE COURSES
MPB***3	English Core 2 Pre- requisite: MPB***3 English Core 1	3	
TOTAL		**24**	

- Malaysian students are REQUIRED to take **MPUK3411 Bahasa Melayu Ilmiah** as a compulsory audit course and will not calculated in CGPA.
- All student must choose ONE (1) co-curricular area of interest, and fulfil the requirement of that chosen area.
- Students are required to take the English Core Courses based on their MUET/IELTS/TOEFL results obtained during the admission application. Refer to **APPENDIX A**

B. CORE COMPUTING (18 CREDIT HOURS)

Students must take ALL courses under core computing courses as follows:

CODE	COURSE	CREDIT	PRE-REQUISITE
STICK1013	Programming Fundamental	3	None
STICK1023	Computer Architecture	3	None
STICK1113	Network and Data Communication	3	STICK1023 Computer Architecture
STICK1123	Operating System	3	STICK1023 Computer Architecture
STICK2013	Database Fundamental	3	STICK1013 Programming Fundamental
STICK2023	System Analysis and Design Fundamentals	3	STICK1013 Programming Fundamental

C. DISCIPLINE CORE (36 CREDIT HOURS)

Students must take ALL courses under discipline core courses as follows:

CODE	COURSE	CREDIT	PRE-REQUISITE
STIQK1123	Discrete Structure	3	None
STIWK1113	Object Oriented Programming	3	STICK1013 Programming Fundamental
STINK2013	Fundamentals of Artificial Intelligence	3	STICK1013 Programming Fundamental
STIWK2013	Data Structures and Algorithm Analysis	3	STIWK1113 Object Oriented Programming

STIWK2023	Software Engineering	3	STICK1013 Programming Fundamental
STIHK2113	Interaction Design	3	None
STIMK2113	IT Project Management	3	STICK2023 System Analysis and Design Fundamentals
STIWK2113	Web Application Development	3	STICK2013 Database Fundamental
STIZK2113	Research and Development	3	STICK2023 System Analysis and Design Fundamentals
STIIK3013	Cloud Computing Technologies and Applications	3	STICK1113 Network and Data Communication
STIWK3013	Mobile Application Development	3	STIWK1113 Object Oriented Programming
STIIK3113	Information Assurance & Security	3	STICK1113 Network and Data Communication

D. FIELD ELECTIVE (18 CREDIT HOURS)

i. Students MUST take **SIX (6)** courses from **FOUR (4**) fields below (18 credits). Students can either choose all courses under **ONE(1)** field or MIX between the fields.

ii. Students who wish to **MIX** courses across different fields can refer to the list of courses offered. Refer to **APPENDIX B.**

1. FIELD ELECTIVE: INFORMATION SYSTEM (18 CREDIT HOURS)

CODE	COURSE	CREDIT	PRE-REQUISITE
STIMK2123	Sustainable Technology Solutions	3	None
STIMK2133	Database Management and Administration	3	STICK2013 Database Fundamental
STIMK3013	Strategic Management of IT	3	STIMK2113 IT Project Management
STIHK3013	User Experience Design	3	STIHK2113 Interaction Design
STIMK3113	IT Strategic Planning	3	STIMK2113 IT Project Management
STIMK3123	Decision Support System	3	STINK2013 Fundamentals of Artificial Intelligence

OR

2. FIELD ELECTIVE: DATA SCIENCE (18 CREDIT HOURS)

CODE	COURSE	CREDIT	PRE-REQUISITE
STIEK2113	Knowledge Discovery in Database	3	STINK2013 Fundamentals of Artificial Intelligence
STIEK2123	Machine Learning	3	STINK2013 Fundamentals of Artificial Intelligence
STIEK3013	Natural Language Processing	3	STINK2013 Fundamentals of Artificial Intelligence
STIEK3023	Business Intelligence	3	STINK2013 Fundamentals of Artificial Intelligence
STIEK3113	Data Exploratory and Visualization	3	STINK2013 Fundamentals of Artificial Intelligence
STIEK3123	Social Data Analytics	3	STINK2013 Fundamentals of Artificial Intelligence

OR

3. FIELD ELECTIVE: COMPUTER SYSTEM AND NETWORK (18 CREDIT HOURS)

CODE	COURSE	CREDIT	PRE-REQUISITE
STIIK2113	Switching and Routing Technology	3	STICK1113 Network and Data Communication
STIIK2123	Server Administration	3	STICK1123 Operating System
STIIK3023	Enterprise Network and Automation	3	STIIK2113 Switching and Routing Technology
STIIK3033	Wireless and Mobile Network	3	STICK1113 Network and Data Communication
STIIK3123	IOT Application Development	3	STIWK3013 Mobile Application Development
STIIK3133	Network Security	3	STIIK3023 Enterprise Network and Automation

OR

4. FIELD ELECTIVE: SOFTWARE DEVELOPMENT (18 CREDIT HOURS)

CODE	COURSE	CREDIT	PRE-REQUISITE
STIWK2123	Data Engineering	3	STIWK2023 Software Engineering
STIWK2133	Software Testing & Quality Assurance	3	STIWK2023 Software Engineering

STIWK3023	Secure Software Development	3	STIWK2023 Software Engineering
STIWK3033	Human Factors Experimental Design	3	STICK2023 System Analysis and Design Fundamentals
STIWK3113	Service Oriented Architecture	3	STIWK2023 Software Engineering
STIWK3123	Real-Time Programming	3	STIWK1113 Object Oriented Programming

OR

5. FIELD ELECTIVE: MIX (18 CREDIT HOURS)

Students can enrol in any combination of **THREE (3)-**credit courses offered under the existing **FOUR (4) Field Electives** to fulfil a minimum of 18 credit hours, subject to the prerequisite requirements of each course. Refer to **APPENDIX B.**

E. FREE ELECTIVE (6 CREDIT HOURS)

Students are REQUIRED to choose **TWO (2)** course offered by **another School**, taking into consideration the following conditions:

i. Must fulfill the prerequisite requirements if the selected course has prerequisites.
ii. Equivalent courses **CANNOT** be taken as a free elective.
iii. University Core Courses, English Core Courses, and Foreign Language Courses **CANNOT** be taken as Free Electives.

F. FINAL YEAR PROJECT (6 CREDIT HOURS)

CODE	COURSE	CREDIT	PRE-REQUISITE
STIZK3993	Academic Project I	3	STIZK2113 Research and Development
STIZK4993	Academic Project II	3	STIZK3993 Academic Project I

H. INDUSTRIAL TRAINING (12 CREDIT HOURS)

i. Student MUST pass all courses in the programme structure before going for industrial training.

ii. Students are required to complete the Industrial Training for SIX (6) months in the final semester.

iii. Student MUST undergo industrial training in organization related to the programme.

CODE	COURSE	CREDIT	PRE-REQUISITE
STIXK49412	Industrial Training	12	Pass all courses in the Programme Structure

PROGRAMME SUGGESTED COURSES BY SEMESTER FOR
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY WITH HONOURS [B.Sc. IT (Hons)]

SEMESTER 1			SEMESTER 2		
Code	**Course**	**Credit Hour**	**Code**	**Course**	**Credit Hour**
MPUK3112/ MPUK3142	Philosophy and Current Issues (Malaysian students) / Malaysian CITRA (International students)	2	MPUK3122 MPUK3132	Malaysian Nation-State Aspiration (Malaysian students) / Basic Malay for Foreigner (International students)	2
MPUK3312	Nationhood Studies	2	MPUK3212	Principles of Entrepreneurship	2
MPB***3	English Core 1	3	MPB***3	English Core 2	3
STICK1013	Programming Fundamental	3	STICK1113	Network and Data Communication	3
STICK1023	Computer Architecture	3	STICK1123	Operating System	3
SQQSK1013	Elementary Statistics	3	STIQK1123	Discrete Structure	3
V******1	Co-curriculum	1	STIWK1113	Object Oriented Programming	3
MPUK3411	Bahasa Melayu Ilmiah	1	V******1	Co-curriculum	1
	TOTAL	**17**		**TOTAL**	**20**
SEMESTER 3			**SEMESTER 4**		
Code	**Course**	**Credit Hour**	**Code**	**Course**	**Credit Hour**
MPUK3152	Integrity and Anti-Corruption	2	STIHK2113	Interaction Design	3
STICK2013	Database Fundamental	3	STIMK2113	IT Project Management	3
STICK2023	System Analysis and Design Fundamentals	3	STIWK2113	Web Application Development	3
STINK2013	Fundamentals of Artificial Intelligence	3	STIZK2113	Research and Development	3
STIWK2013	Data Structures and Algorithm Analysis	3	STI*K21*3	Field Elective Course 1	3
STIWK2023	Software Engineering	3	STI*K21*3	Field Elective Course 2	3
	TOTAL	**17**		**TOTAL**	**18**
SEMESTER 5			**SEMESTER 6**		
Code	**Course**	**Credit Hour**	**Code**	**Course**	**Credit Hour**
STIIK3013	Cloud Computing Technologies and Applications	3	BWFFK1013	Fundamentals of Finance	3
STIWK3013	Mobile Application Development	3	STIIK3113	Information Assurance & Security	3
STIZK3993	Academic Project I	3	STIZK4993	Academic Project II	3
*******3	Free Elective 1	3	*******3	Free Elective 2	3
STI*K30*3	Field Elective Course 3	3	STI*K31*3	Field Elective Course 5	3
STI*K30*3	Field Elective Course 4	3	STI*K31*3	Field Elective Course 6	3
	TOTAL	**18**		**TOTAL**	**18**
SEMESTER 7			**OVERALL TOTAL**		
Code	**Course**	**Credit Hour**			
STIXK49412	Industrial Training	12	**120**		
	TOTAL	**12**			

APPENDIX B

LIST OF COURSES FOR FIELD ELECTIVE: OPEN

NO.	CODE	COURSE	CREDIT	PRE-REQUISITE
1.	STIMK2123	Sustainable Technology Solutions	3	None
2.	STIMK2133	Database Management and Administration	3	STICK2013 Database Fundamental
3.	STIMK3013	Strategic Management of IT	3	STIMK2113 IT Project Management
4.	STIHK3013	User Experience Design	3	STIHK2113 Interaction Design
5.	STIMK3113	IT Strategic Planning	3	STIMK2113 IT Project Management
6.	STIMK3123	Decision Support System	3	STINK2013 Fundamentals of Artificial Intelligence
7.	STIEK2113	Knowledge Discovery in Database	3	STINK2013 Fundamentals of Artificial Intelligence
8.	STIEK2123	Machine Learning	3	STINK2013 Fundamentals of Artificial Intelligence
9.	STIEK3013	Natural Language Processing	3	STINK2013 Fundamentals of Artificial Intelligence
10.	STIEK3023	Business Intelligence	3	STINK2013 Fundamentals of Artificial Intelligence
11.	STIEK3113	Data Exploratory and Visualization	3	STINK2013 Fundamentals of Artificial Intelligence
12.	STIEK3123	Social Data Analytics	3	STINK2013 Fundamentals of Artificial Intelligence
13.	STIIK2113	Switching and Routing Technology	3	STICK1113 Network and Data Communication
14.	STIIK2123	Server Administration	3	STICK1123 Operating System
15.	STIIK3023	Enterprise Network and Automation	3	STIIK2113 Switching and Routing Technology
16.	STIIK3033	Wireless and Mobile Network	3	STICK1113 Network and Data Communication
17.	STIIK3123	IOT Application Development	3	STIWK3013 Mobile Application Development
18.	STIIK3133	Network Security	3	STIIK3023 Enterprise Network and Automation
19.	STIWK2123	Data Engineering	3	STIWK2023 Software Engineering
20.	STIWK2133	Software Testing & Quality Assurance	3	STIWK2023 Software Engineering
21.	STIWK3023	Secure Software Development	3	STIWK2023 Software Engineering
22.	STIWK3033	Human Factors Experimental Design	3	STICK2023 System Analysis and Design Fundamentals
23.	STIWK3113	Service Oriented Architecture	3	STIWK2023 Software Engineering
24.	STIWK3123	Real-Time Programming	3	STIWK1113 Object Oriented Programming

BACHELOR OF COMPUTER SCIENCE WITH HONOURS [B.Comp.Sc. (Hons)]

The Bachelor of Computer Science with Honours program consists of eight (8) components as follows:

COMPONENTS	CREDIT HOURS
A. UNIVERSITI CORE	12
B. ENGLISH CORE	6
C. PROGRAM CORE	18
D. DISCPLINE CORE	33
E. PROGRAM ELECTIVES	33
F. FREE ELECTIVES	3
G. ACADEMIC PROJECT	6
H. INDUSTRIAL TRAINING	12
TOTAL	**123**

To be awarded the Bachelor of Computer Science with Honours, students must complete a minimum of 123 credit hours according to the following structure:

A. UNIVERSITY CORE COURSES (12 CREDIT HOURS)

i. Malaysian Students

CODE	COURSE	CREDIT HOURS
MPUK3112	Falsafah dan Cabaran Semasa	2
MPUK3122	Malaysian Nation-State Aspiration	2
MPUK3312	Nationhood Studies	2
MPUK3212	Principles of Entrepreneurship	2
MPUK3152	Integrity And Anti- Corruption	2
MPUK3411	Bahasa Melayu Ilmiah **audit course*	1
VXXXXXXX	Co-Curriculum	2
	TOTAL	**13**

ii. International Students

CODE	COURSE	CREDIT HOURS
MPUK3142	Malaysian CITRA	2
MPUK3132	Basic Malay for Foreigner	2
MPUK3312	Nationhood Studies	2
MPUK3212	Principles of Entrepreneurship	2
MPUK3152	Integrity And Anti- Corruption	2
VXXXXXXX	Co-Curriculum	2
	TOTAL	**12**

- Malaysian students are **REQUIRED** to take **MPUK3411 Bahasa Melayu Ilmiah** as a compulsory audit course and will not calculated in CGPA.
- All students must choose ONE (1) co-curricular area of interest, and fulfill the requirements of that chosen area.

B. ENGLISH CORE COURSES (6 CREDIT HOURS)

Students are required to take the English Core Courses based on their MUET/IELTS/TOEFL results obtained during the admission application. Refer to **APPENDIX A**

C. PROGRAM CORE (18 CREDIT HOURS)

CODE	COURSE NAME	CREDIT	PREREQUISITE
STICK1013	Programming Fundamental	3	None
STICK1023	Computer Architecture	3	None
STICK1113	Network and Data Communication	3	None
STICK1123	Operating System	3	None
STICK2013	Database Fundamental	3	None
STICK2023	System Analysis and Design Fundamentals	3	None

D. DISCPLINE CORE (33 CREDIT HOURS)

CODE	COURSE NAME	CREDIT	PREREQUISITE
STTHK1013	Interaction Design	3	None
STTQK1113	Discrete Mathematics	3	None
STTPK1113	Intermediate Programming	3	STICK1013 Programming Fundamental
STTPK2013	Data Structures	3	STTPK1113 Intermediate Programming
STTPK2023	Advanced Programming	3	STTPK1113 Intermediate Programming
STTNK2013	Artificial Intelligence	3	None
STTPK2113	Algorithm Design and Analysis	3	STTPK2013 Data Structures
STTHK2113	Data Analytics	3	STTNK2013 Artificial Intelligence
STTPK2033	Software Development Process	3	None
STTPK2123	Concepts and Paradigms in Programming Language	3	STTPK1113 Intermediate Programming
STTCK3033	Secure Software Development	3	STTPK2033 Software Development Process

E. PROGRAM ELECTIVES (33 CREDIT HOURS)

Students must select **ONE (1)** Elective Area from the following **TWO (2)** Elective Areas:

1. **ELECTIVE AREA CYBERSECURITY**

CODE	COURSE NAME	CREDIT	PREREQUISITE
STTCK2113	Digital World Security	3	STICK1113 Network and Data Communication

STTCK2123	Cryptography	3	STTQK1113 Discrete Mathematics
STTCK2013	Security Architecture	3	STICK1113 Network and Data Communication
STTCK3013	Policy and Access Control	3	STTCK2113 Digital World Security
STTCK3023	Cybersecurity Risk Management	3	STTCK2113 Digital World Security
STTCK3113	Ethical Issues in Cybersecurity	3	None
STTCK3123	Digital Forensics	3	STTCK2113 Digital World Security
STTCK3133	Malicious Threats	3	STTCK2113 Digital World Security
STTCK3143	Cyber Conflicts	3	STTCK2113 Digital World Security
STTJK2113	Routing and Switching	3	STICK1113 Network and Data Communication
STTJK3013	Network Scaling and Connectivity	3	STTJK2113 Routing and Switching

2. ELECTIVE AREA HUMAN-CENTERED COMPUTING

CODE	COURSE NAME	CREDIT	PREREQUISITE
STTHK2123	System and Tool Interaction	3	None
STTHK2133	Modelling and Simulation	3	None
STTGK3013	Mobile Web Programming	3	None
STTHK3013	Pattern Recognition and Analysis	3	STTHK2123 System and Tool Interaction
STTHK3023	Collaboration and Social Computing	3	STTHK2113 Data Analytics
STTHK3033	Information Visualization	3	STTHK2113 Data Analytics
STTHK3113	Sensor-Based Systems	3	STTPK2023 Advanced Programming
STTHK3123	Human Factors Experimental Design	3	None
STTHK3133	Human Factors Engineering	3	None
STTPK3023	Software Testing	3	None
STTPK3123	Software Quality Management	3	None

F. FREE ELECTIVES (3 CREDIT HOURS)

Students are REQUIRED to choose **ONE (1)** course offered by **another School**, taking into consideration the following conditions:

i. Must fulfill the prerequisite requirements if the selected course has prerequisites.
ii. Equivalent courses **CANNOT** be taken as a free elective.
iii. University Core Courses, English Core Courses, and Foreign Language Courses **CANNOT** be taken as Free Electives.

G. ACADEMIC PROJECT (6 CREDIT HOURS)

CODE	COURSE NAME	CREDIT	PREREQUISITE
STTZK3993	Academic Project I	3	STTPK2023 Advanced Programming
STTZK4993	Academic Project II	3	STTZK3993 Academic Project I

H. INDUSTRIAL TRAINING (12 CREDIT HOURS)

Students are **REQUIRED TO PASS** all courses in the programme structure **before undergoing Industrial Training.**

CODE	COURSE NAME	CREDIT	PREREQUISITE
STTXK49512	Industrial Training	12	STTZK4993 Academic Project II

RECOMMENDED COURSE ENROLMENT BY SEMESTER FOR THE BACHELOR OF COMPUTER SCIENCE WITH HONOURS PROGRAMME

I. ELECTIVE AREA CYBERSECURITY

SEMESTER 1			SEMESTER 2		
Code	**Course Name**	**Credit**	**Code**	**Course Name**	**Credit**
MPBXXXX / SBLEKXXXX	English Core	3	MPBXXXX / SBLEKXXXX	English Core	3
STTHK1013	Interaction Design	3	MPUK3152	Integrity and Anti-Corruption	2
STICK1113	Network and Data Communication	3	STICK2023	System Analysis and Design Fundamentals	3
STTQK1113	Discrete Mathematics	3	STICK2013	Database Fundamental	3
STICK1023	Computer Architecture	3	STICK1123	Operating System	3
STICK1013	Programming Fundamental	3	STTPK1113	Intermediate Programming	3
VXXXXX1	Co-Curriculum	1	VXXXXXX1	Co-Curriculum	1
TOTAL		**19**	**TOTAL**		**18**
SEMESTER 3			**SEMESTER 4**		
Code	**Course Name**	**Credit**	**Code**	**Course Name**	**Credit**
MPUK3122 / **MPUK3132	Malaysian Nation-State Aspiration/ **Basic Malay for Foreigner	2	MPUK1042	Nationhood Studies	2
STTPK2013	Data Structures	3	STTPK2113	Algorithm Design and Analysis	3
STTPK2033	Software Development Process	3	STTCK3033	Secure Software Development	3
STTNK2013	Artificial Intelligence	3	STTCK2123	Cryptography	3
STTCK2113	Digital World Security	3	STTHK2113	Data Analytics	3
STTPK2023	Advanced Programming	3	STTPK2123	Concepts and Paradigms in Programming Language	3
XXXXXXX3	Free Elective	3	STTJK2113	Routing and Switching	3
TOTAL		**20**	**TOTAL**		**20**
SEMESTER 5			**SEMESTER 6**		
Code	**Course Name**	**Credit**	**Code**	**Course Name**	**Credit**
MPUK3212	Principles of Entrepreneurship	2	MPUK3112 / **MPUK3142	Falsafah dan Cabaran Semasa / **Malaysian CITRA	2
STTCK2013	Security Architecture	3	STTCK3113	Issues in Cybersecurity	3
STTCK3013	Policy and Access Control	3	STTCK3123	Digital Forensics	3
STTCK3023	Cybersecurity Risk Management	3	STTCK3133	Malicious Threats	3
STTJK3013	Network Scaling and Connectivity	3	STTCK3143	Cyber Conflicts	3
STTZK3993	Academic Project I	3	STTZK4993	Academic Project II	3
TOTAL		**17**	**TOTAL**		**17**
SEMESTER 7			**GRAND TOTAL**		
Code	**Course**	**Credit**			
STTXK49512	Industrial Training	12	**123**		
TOTAL		**12**			

II. ELECTIVE AREA HUMAN CENTERED COMPUTING

SEMESTER 1			SEMESTER 2		
Code	**Course Name**	**Credit**	**Code**	**Course Name**	**Credit**
MPBXXXX / SBLEKXXXX	English Core	3	MPBXXXX / SBLEKXXXX	English Core	3
STTHK1013	Interaction Design	3	MPUK3152	Integrity and Anti-Corruption	2
STICK1113	Network and Data Communication	3	STICK2023	System Analysis and Design Fundamentals	3
STTQK1113	Discrete Mathematics	3	STICK2013	Database Fundamental	3
STICK1023	Computer Architecture	3	STICK1123	Operating System	3
STICK1013	Programming Fundamental	3	STTPK1113	Intermediate Programming	3
VXXXXX1	Co-Curriculum	1	VXXXXXX1	Co-Curriculum	1
TOTAL		**19**	**TOTAL**		**18**
SEMESTER 3			**SEMESTER 4**		
Code	**Course Name**	**Credit**	**Code**	**Course Name**	**Credit**
MPUK3122 / **MPUK3132	Malaysian Nation-State Aspiration/ **Basic Malay for Foreigner	2	MPUK1042	Nationhood Studies	2
STTPK2013	Data Structures	3	STTPK2113	Algorithm Design and Analysis	3
STTPK2033	Software Development Process	3	STTHK2113	Data Analytics	3
STTNK2013	Artificial Intelligence	3	STTCK3033	Secure Software Development	3
STTHK2123	System and Tool Interaction	3	STTHK2133	Modelling and Simulation	3
STTPK2023	Advanced Programming	3	STTPK2123	Concepts and Paradigms in Programming Language	3
XXXXXXX3	Free Elective	3	STTPK3023	Software Testing	3
TOTAL		**20**	**TOTAL**		**20**
SEMESTER 5			**SEMESTER 6**		
Code	**Course Name**	**Credit**	**Code**	**Course Name**	**Credit**
MPUK3212	Principles of Entrepreneurship	2	MPUK3112 / **MPUK3142	Falsafah dan Cabaran Semasa / **Malaysian CITRA	2
STTGK3013	Mobile Web Programming	3	STTHK3033	Information Visualization	3
STTPK3123	Software Quality Management	3	STTHK3113	Sensor-Based Systems	3
STTHK3013	Pattern Recognition and Analysis	3	STTHK3023	Collaboration and Social Computing	3
STTHK3123	Human Factors Experimental Design	3	STTHK3133	Human Factors Engineering	3
STTZK3993	Academic Project I	3	STTZK4993	Academic Project II	3
TOTAL		**17**	**TOTAL**		**17**
SEMESTER 7			**GRAND TOTAL**		
Code	**Course**	**Credit**	**123**		
STTXK49512	Industrial Training	12			
TOTAL		**12**			

APPENDIX A

ENGLISH CORE COURSES

A student is required to take English core courses according to these requirements:

PATH (TOTAL CREDIT HOURS) AND COURSES SHOULD BE TAKEN	PRE REQUISITE	MUET	IELTS	TOEFL (iBT)	PTE ACADEMIC	CEFR
PATH 1 (9 CREDIT HOURS) • MPB1013 BASIC ENGLISH PROFICIENCY (credit bearing) • MPB2013 ENGLISH LANGUAGE ENRICHMENT I • MPB3013 ENGLISH LANGUAGE ENRICHMENT II	Have taken English Intensive course for international students	BAND 1.0 BAND 2.5	<4.0	0 - 30	0 - 29	A2
PATH 2 (6 CREDIT HOURS) • MPB2013 ENGLISH LANGUAGE ENRICHMENT I • MPB3013 ENGLISH LANGUAGE ENRICHMENT II	Have taken MPB1013 Basic English Proficiency	BAND 3.0 BAND 3.5	4.0 - 5.0	30 - 45	30 - 41	B1 - B1+
PATH 3 (6 CREDITS HOURS) • MPB3013 ENGLISH LANGUAGE ENRICHMENT II • **1 ESP COURSE**	Have taken MPB2013 English Language Enrichment I	BAND 4.5	6.0 – 6.5	60-93	50-63	B2
		BAND 4.0	5.5	46 - 59	42 - 49	B2+
LIST OF ESP COURSES: • SBLEK3023 English for Small Group Communication • SBLEK3033 Report Writing • SBLEK3043 Hospitality English • SBLEK3053 Public Speaking Skills • SBLEK3063 English for Professional Communication	Have taken MPB3013 English Language Enrichment II					
EXEMPTED FROM ANY ENGLISH LANGUAGE COURSES *except for language courses that are listed as programme core courses		BAND 5.0	7.0 - 8.0	94 - 114	64 - 82	C1